Analog IC Design Implementation and Verification

Weeks 12 and 13 – Build Layouts Using Matching and Shielding Techniques

Version IC 23.1 ISR11.29 with SPECTRE 24.1 ISR1

Revision 1.0

Estimated time: 1 Week

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Build Basic Layouts from the Schematic

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Module Objectives

In this module, you learn how to

- Identify good commercial design practices for building layouts from schematics.
- Annotate a schematic with appropriate layout constraints.
- Interpret a constrained schematic and specify the layout techniques that are needed to meet those constraints.

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Process and Layout Variations

Fabrication of ICs is a combination of different steps on silicon.

- These steps are not perfect and have some defects.
- Defects cause process variations and the fabricated circuits do not behave like ideal circuits.
- Knowing these variations helps you to design more robust circuits that are not affected by the variations.

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Causes of Variations

There are different causes of variations in chips:

- Process variation
  - Doping and ion implantation
  - Oxidation
  - Etching
- Temperature variation
- Layout variation

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Doping and Ion Implantation

Doping and ion implantation are used to build different devices.

- For example, they are used to create an N+ region to build drain and source of NMOS.
- Doping (n-type or p-type) might vary across the surface and the depth of the wafer.
- Like deposition, ion implantation has variation across the wafer. However, it can be controlled better than deposition.
- Ion implantation causes damage to the silicon crystal structure.
- Thermal annealing is a process after doping or after ion implantation that is used to change the doping profile or to get more uniform implantation.

  This step also varies from wafer to wafer because of the heat transfer and time of annealing.

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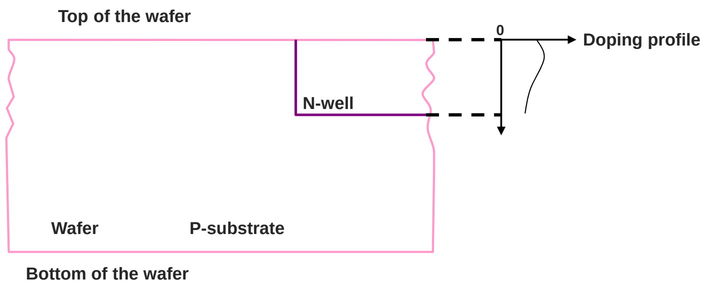
Doping and Ion Implantation (continued)



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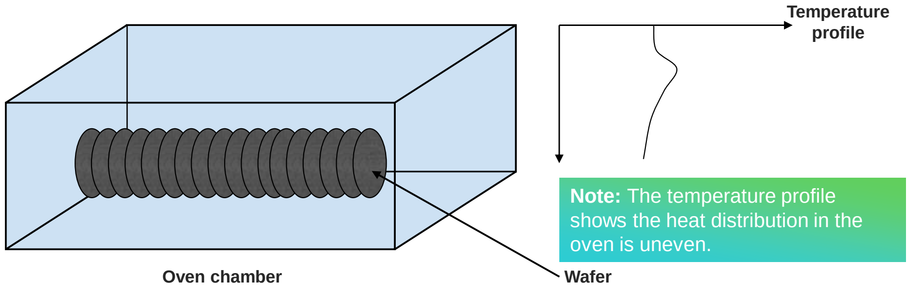
Oxidation

Oxidation is used to grow a thin oxide layer on the surface of the wafer. The thickness of the oxide is one of the most important factors in the specification of PMOS or NMOS devices. The thickness depends on several factors.

- Temperature and the oxygen availability at the site of oxidation.
  - These dependencies cause variations.
- Impurities during oxidation.



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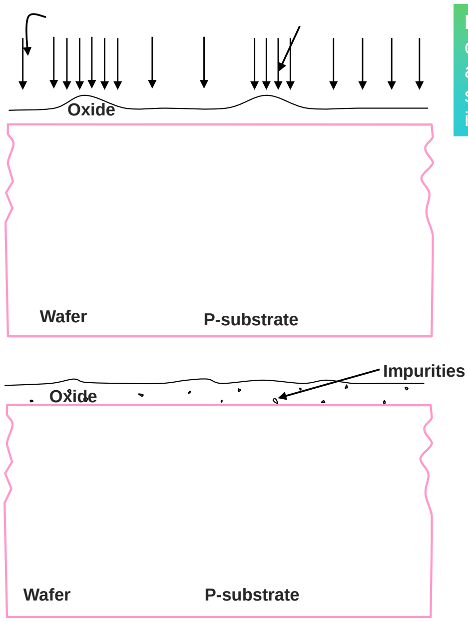
Oxidation (continued)

Arrows show the available oxygen at the surface of the silicon wafer.



Note: Where more oxygen is available at the surface of the silicon wafer, the oxide is thicker.

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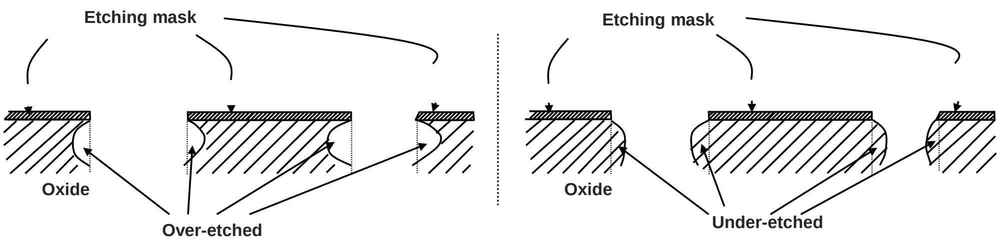
Etching

To make devices, some layers, such as poly, need to be etched to build a gate of a transistor. The size of the gate (W and L) determines the specification of the NMOS or PMOS.

- There are two kinds of etching:
  - Dry etching: Material is sputtered or dissolved using reactive ions or a vapor phase etchant.
  - Wet etching: The material is dissolved when immersed in a chemical solution.
- During the etching process, under-etch or over-etch can occur. These affect the performance of the device.



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Temperature Variation

All semiconductor devices have temperature dependency. Some devices have less and some more dependency.

- This variation results from the effect of temperature on the parameters of the device.

  For example, the mobility of electrons and the threshold voltage of MOS will change with temperature and that will affect the current of a device with fixed Vgs.

- The circuit designer should consider temperature variations during circuit design, so the circuit will be protected from them.
- The layout designer should be concerned about the temperature but more concerned about the stress that higher temperature puts on die. This concern will be discussed later in the subsection on thermal issues.

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Layout Variations

Chip layout introduces variations.

- Photolithography usually determines the minimum transistor size for a process.
  However, the minimum dimension does not mean minimum variation of the process.
  - For example, a 0.35 µm X 0.35 µm poly resistor will have greater variation than a 10 µm x 10 µm poly resistor.
  - The larger the area of a device, the more that process variation averages out.
- Different blocks have different functionalities, so their circuits and layout are different.
- The difference in layout results in different layout patterns on the top level.
- This difference in layout puts stress on different locations on the die.

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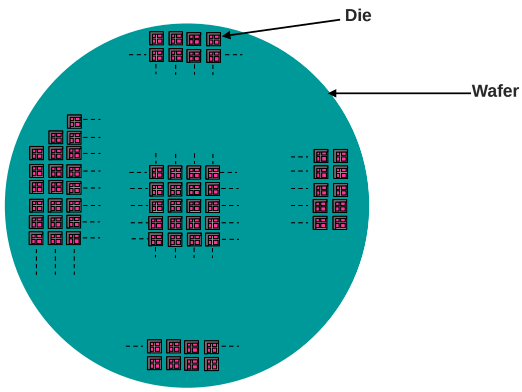
Location of Variation

- Based on the size of the die and wafer, there can be hundreds of dies on each wafer.
- Die layout repeats over the wafer to fill out the wafer area.



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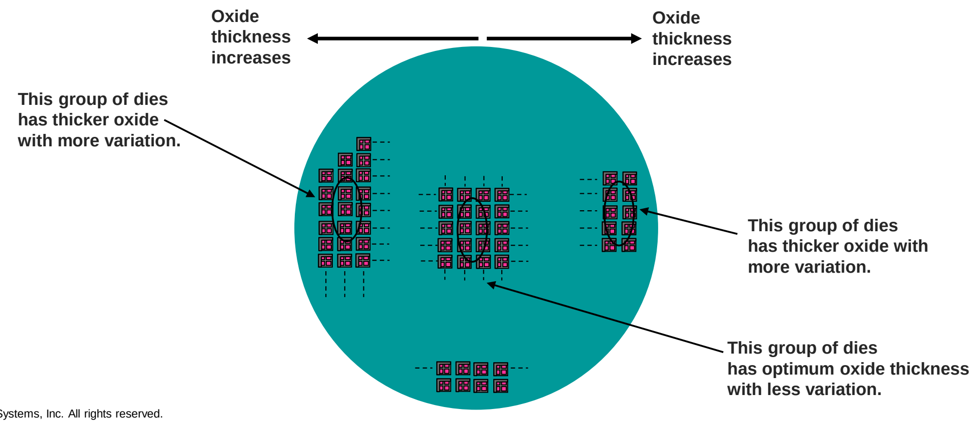
Location of Variation (continued)

- Process variation happens on the wafer at specific angles. For example, the gradient of oxide-thickness variation will happen in a certain angle, depending on the temperature profile and available oxygen in the chamber.
- Usually center of the wafer is the best point for reducing process variation. The dies that are close to the center of wafer have less variation than the ones close to the edge.



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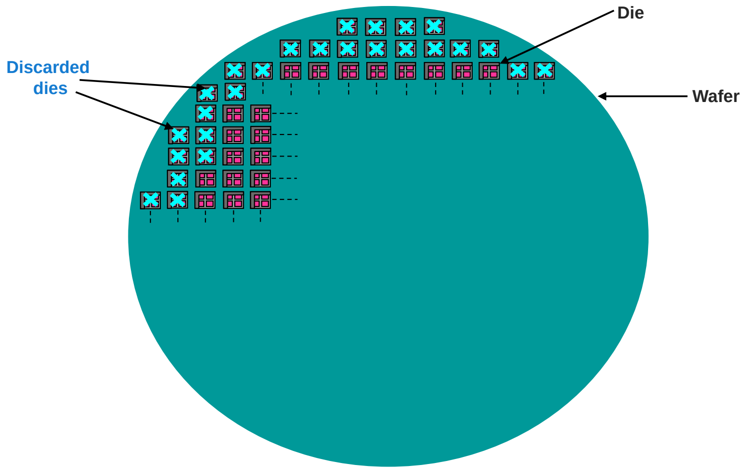
Location of Variation (continued)

Dies that are very close to the edge of the wafer are discarded usually because of higher variation of process parameters.



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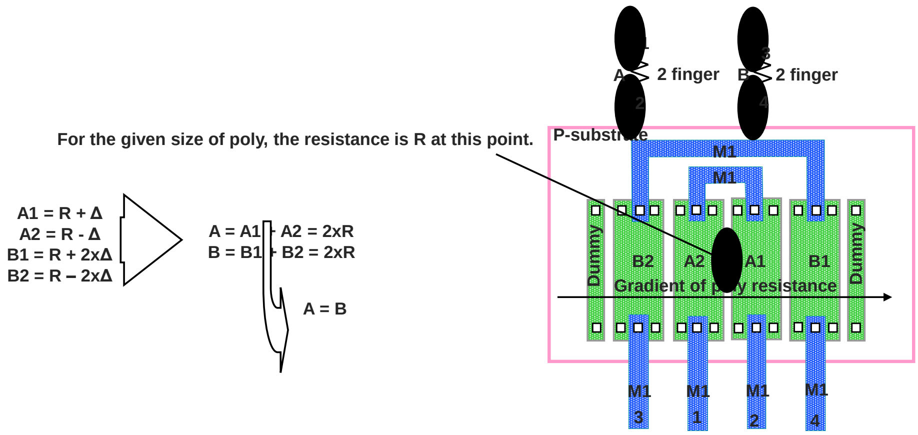
Location of Variation (continued)

- The gradient of these variations will be in a certain angle from one side of the die to the opposite side of the die.
- Having common-centroid matching will cancel the mismatch between any two or more devices. However, the absolute value of the parameters will change based on the gradient of variation.



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Location of Variation (continued)

- The gradient of these variations will be in a certain angle from one side of the die to the opposite side of the die.
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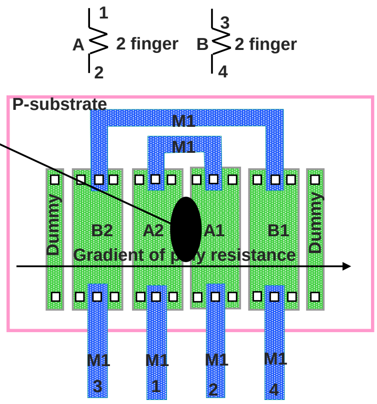
For the given size of poly, the resistance is R at this point.

A1 = R + Δ
A2 = R - Δ
B1 = R + 2xΔ
B2 = R – 2xΔ

⇒

A = A1 + A2 = 2xR
B = B1 + B2 = 2xR

A = B



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Effect of Variation

- Effects of these variations are in the parameters of the devices. For example:
  - W and L of a poly resistor can be changed because the etching variation over the die causes different poly resistance in different locations of the die although the layouts are identical.
  - Differences in the oxide thickness causes different Cox (which is important parameter for current equation of transistor) for identical CMOS devices.
- All these variations will affect the overall performance of the die.

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How to Overcome These Variations

The first and most important step a layout designer takes to overcome process and layout variations is to understand and determine the sensitive devices and matching required devices on each block, the current consumption of the block, hot devices in each block, and the location of the block in floorplan of the whole die.

- Based on this information, which usually is supplied by a circuit designer, a layout designer can build a floorplan for each block based on the inputs and outputs of the block.
- In this floorplan the matching requirement, width of the connections, dummy devices, orientation of devices, and other layout constraints should be considered.

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The assumption is that circuit designer has designed his/her circuit in optimum way based on the information which is available from each foundry . For example if he/she needs to have very linear resistor he/she should not use Nwell resistor because of high voltage coefficient of it and instead he/she should use poly resistors. Or circuit designer based on the matching which is required for example between two resistor should chose correct L and W for each finger of these two resistor to meet the matching requirement.

Guidelines for Overcoming Variations

Circuit designer

- Design using matching of devices instead of absolute value.
- Choose the available devices according to the functionality. For example, do not use well resistors (known for high-voltage coefficient) if voltage range involved is big and absolute value is important.
- Increase the size of devices (with area and speed considerations) to minimize length and width variations.
- Choose the best finger number for matched devices to share the contacts.

Layout designer

- Use common-centroid matching for sensitive devices, such as input-differential pairs of amplifiers, current-mirror transistors or matched-required resistors
- Use redundancy in contacts and vias because of IR drop and reliability concerns.
- Use the same orientation for the gates of transistors
- Choose appropriate width for the metal connections based on the current.

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Matching and Shielding Techniques

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Subsection

Layout Constraints

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Layout Constraints

Layout constraints are limiting factors that the analog-layout designer should know to create a more accurate layout.

- The basic operations of some circuits directly depend on layout constraints, such as, the offset or CMRR for a differential amplifier.
- Unlike digital circuits, the performance of an analog circuit depends on these constraints.
- Usually, these constraints should be considered for specific sets of devices in complicated circuits and the circuit designer determines this specific set.

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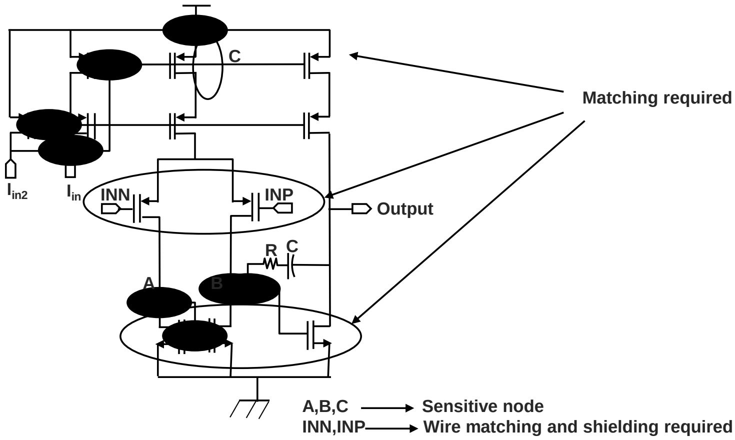
Layout Constraints (continued)

For example, in the schematic diagram below, not all the devices need to be matched. Also, specific nodes are sensitive and should be considered carefully.



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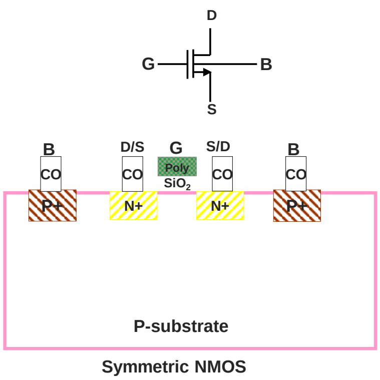
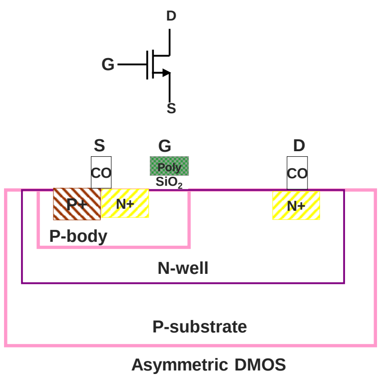
Symmetry

From the viewpoint of geometry, there are two kinds of devices.

- **Symmetric devices,** such as NMOS and PMOS where you can swap drain and source
- **Asymmetric devices,** such as DMOS where you cannot swap drain and source



Symmetric NMOS



Asymmetric DMOS

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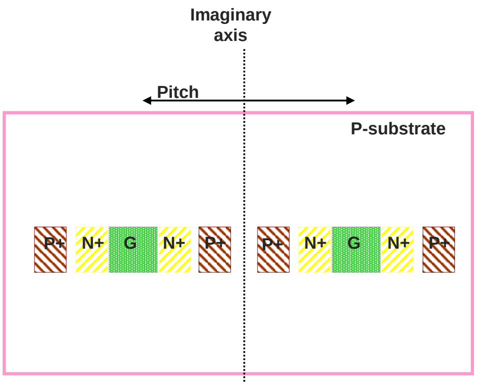
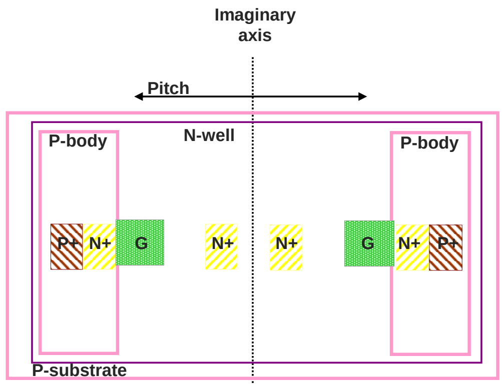
Symmetry (continued)

- Two symmetric devices should have same orientation and should be a mirror of each other with respect to imaginary axes.
- Both symmetric and asymmetric devices can have a symmetric layout.



Symmetric layout of symmetric device



Symmetric layout of asymmetric device

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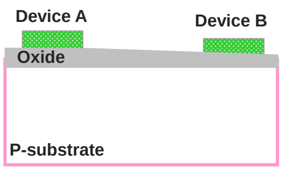
Matching

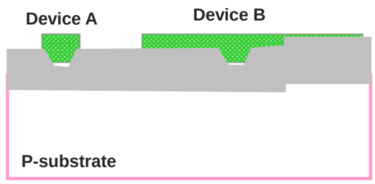
The goal of matching is to obtain close specifications between two or more devices. Matched devices have close specifications.

- The important factors in matching are
  - The physical location of devices should be near each other
  - The sizes of devices should be similar

    In transistors this will be the gate area of transistors.
- For example, assume that the oxide thickness varies over the die.



Because of the location difference, device A has thicker oxide below it than device B.



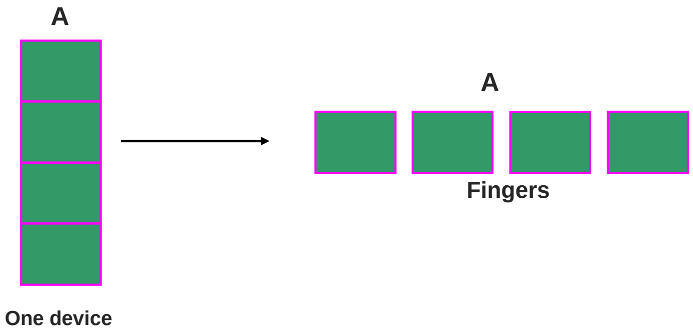
Device A has very small size, so any small variation (such as oxide thickness) will affect its specification. For device B the variations will be averaged out.

Physical location of the device is important because of the variations which are there over the wafer and die itself. If two devices sits far away from each other there will not be any correlation between the effects which these two device are going to see. Then the specification of these two devices will be different.

Most of the electrical specification of the transistors come from the gate oxide and channel specification like gate oxide thickness and L and W so the matching of the transistors will be related to the gate area. The matching of other passive elements like resistors and capacitor is related to the area of those devices

Fingers

For better matching, each device should be broken into multiple pieces to be spread in a larger area. Each of these pieces is called a *finger*.



The whole idea to have common centroid is based on the fact that if there are some variation on the device parameters like oxide thickness or doping gradients, putting fingers centroid will average out these variation between the matched devices.

The negative side of common centroid are higher parasitic routing capacitance, and also complicated placement in higher number of required matched devices.

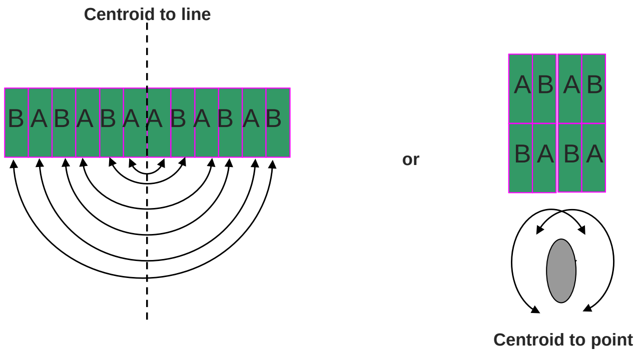
Common centroid techniques is best technique to have the best matching between two devices.

Types of Matching

: Fingers of matched devices are centroid to same line or point.

For a common centroid, there should be an even number of fingers.

In the diagrams, A and B are two devices.



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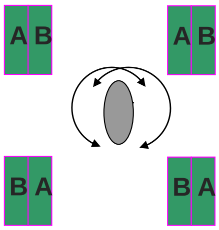
Types of Matching (continued)

Cross quad: This kind of matching is like having a common centroid to a point. The transistors are placed in a square in a bigger area.



Cross quad

Cross Squad is almost same as having centroid to point. The difference is the area which device gets on the chip. And remember more area means better matching. So matching wise this is better than Cross Squad but area wise common centroid is better.

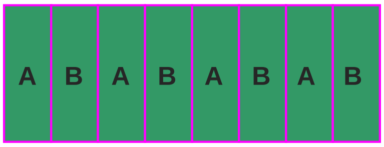
If you try to put other circuits at the center, then coupling noise can be an issue for Cross Squad.

Types of Matching (continued)

Interdigitation is distributing devices in a larger area and sharing every part of this area between two or more devices.

In the diagram, A and B are two devices.

- Interdigitation is not as good as common-centroid matching.
- It is less complicated than common-centroid matching.
- It has less parasitic routing capacitance because of the easier routing.



Interdigitation

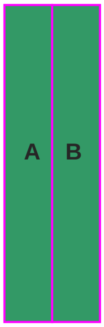
In interdigitation there is no symmetry for the fingers, but they just sit alternately between each other. The idea behind this is just to distribute the device in more area and share every piece of this area between these tow devices.

Types of Matching (continued)

Matching can be done by putting devices closer together.

In the diagram, A and B are two devices.

- This is the weakest kind of matching.
- It does not need to have multiple fingers.

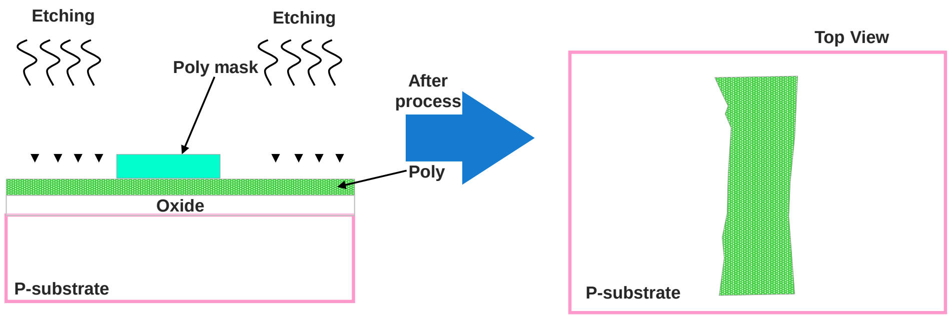


Matching devices

This is the worst kind of matching. It just ensures that there is not a lot of variation in the doping or oxide thickness. It does not need multiple fingers for device. but the amount of matching which you can get will not be good enough for high performance analog ICs.

Dummy Devices

- Etching is an action that happens in the fabrication of all semiconductor devices. It can be the etching of metal or poly.
- This chemical action has some effect on the edge of the exposed area. For example, after etching the poly, the edge is not smooth.



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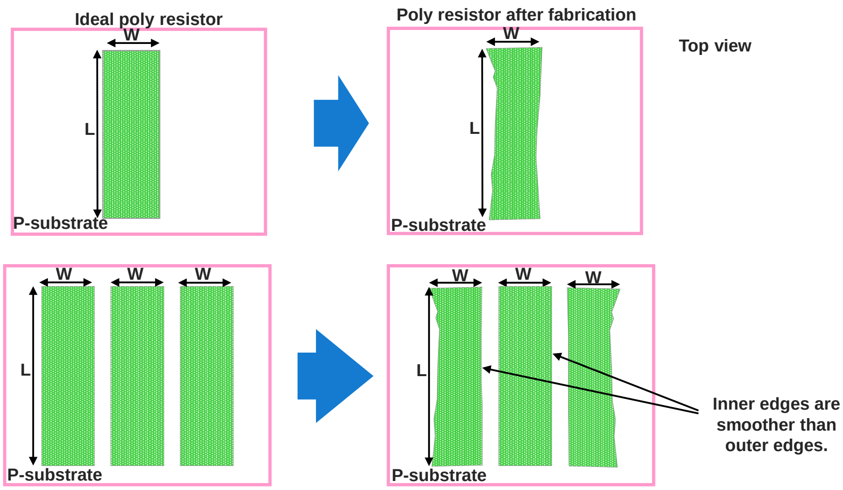
Dummy Devices (continued)

- The uneven edges cause some differences between real devices and ideal devices.
- The effect of over-etching depends on the size of the exposed area.



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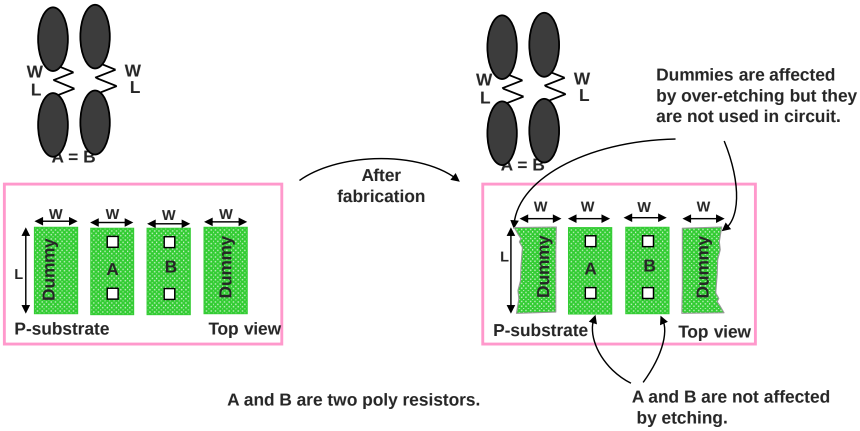
Dummy Devices (continued)

- A *dummy device* is an unused device that is sacrificed to get better main devices.
- A main device is one that acts in the functionality of the circuit.



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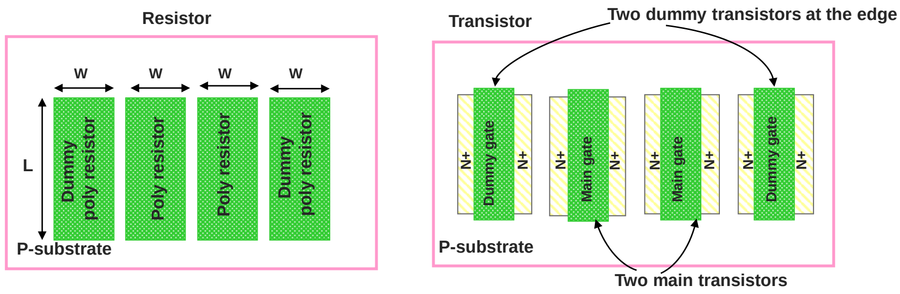
Dummy Devices (continued)

Most devices need etching during fabrication, so there are dummies of most devices such as dummy resistors, dummy transistors, and dummy capacitors.



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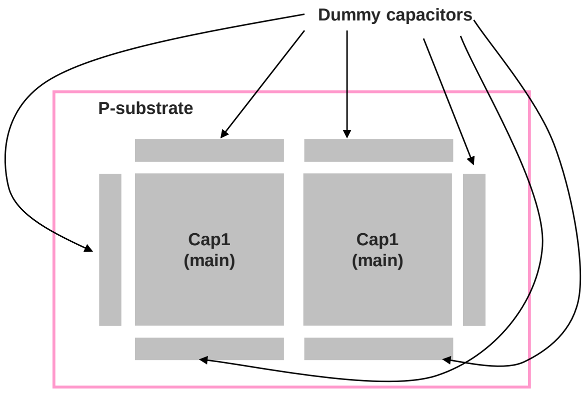
Dummy Devices (continued)

Dummy capacitors



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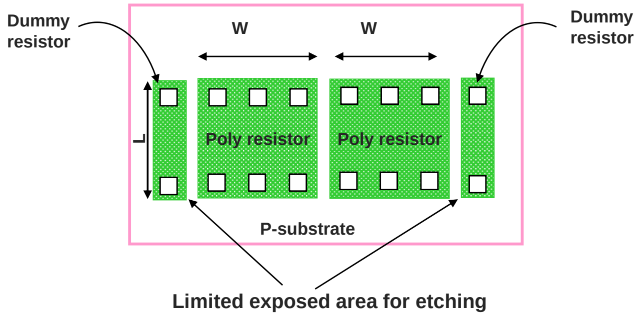
Dummy Devices (continued)

- The only purpose of dummy devices is to limit the exposed area for etching between devices.
- The dummy device can be a minimum size.



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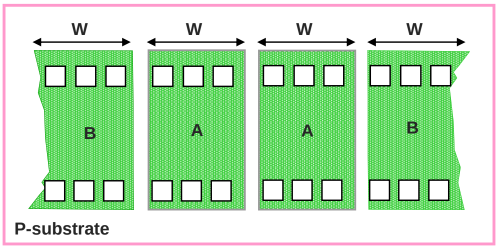
Dummy Devices (continued)

- The etching effect should be same for matched devices.

  A and B are common-centroid-matched resistors, with two fingers each.



Resistor A: Sits at the middle and does not get over-etching effect.

Resistor B: Sits at the beginning and end of the chain and gets the over-etching effect.

The resistor value of A is not same as B because edges of B are not as smooth as A.

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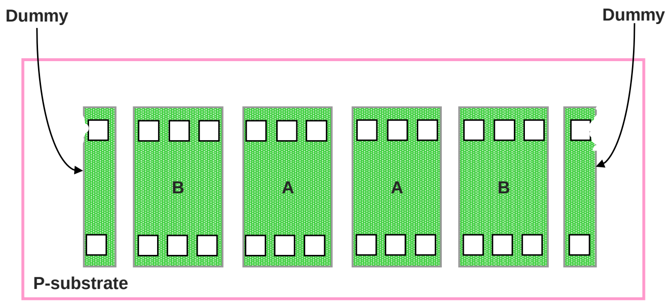
Dummy Devices (continued)

Using dummy devices is one of the most important techniques to get better matching between devices.



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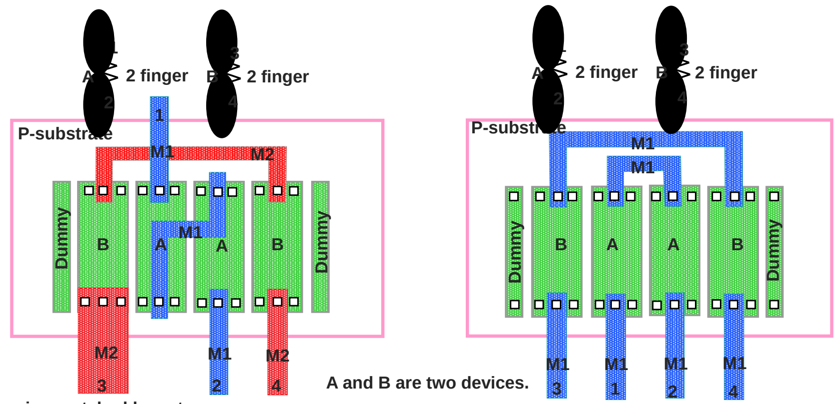
Wire Matching

- After matching two devices, match the connections to those devices also. This is known as net matching or wire matching.
- In differential circuits, wire matching can be as important as device matching.
- In wire matching, the width and the length of the wire are important and must be the same for differential signals.



No wire- matched layout

Wire- matched layout

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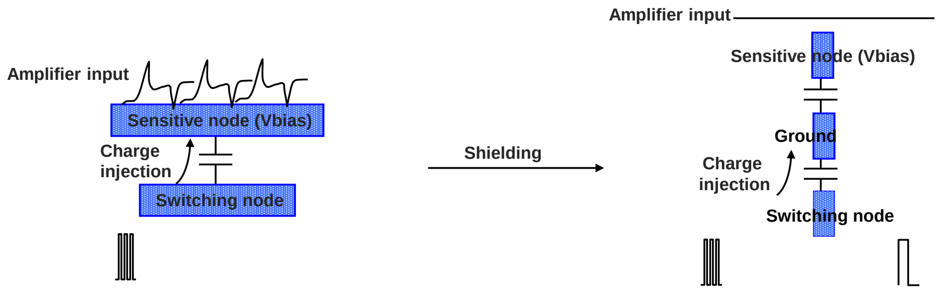
Shielding

Noise is one of the major concerns in analog designs.

- Coupling noise is one of the important noise sources that is related to the layout.
- Coupling noise is caused by the parasitic capacitance of the layout.

Shielding is a technique to reduce parasitic capacitance between a noisy node, such as switching, and a sensitive node, such as an amplifier input.

- Connecting a shield line to the ground takes the coupled charge to the ground and prevents it from coupling to the sensitive node.



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Shielding (continued)

- Amplifier outputs and bias voltages are low-impedance nodes, so theoretically they should be immune to charge injection.
  - However, at the frequency at which charge is being injected, the impedance of routing from these outputs to the coupling point degrades this immunity.
  - Shielding should be considered for these nodes also.
- Shielding every single line separately takes lots of space on layout.
  - It is better to put all sensitive and analog signals in a group and then shield them.
  - For example, the inputs of differential amplifier can be shielded in a group.

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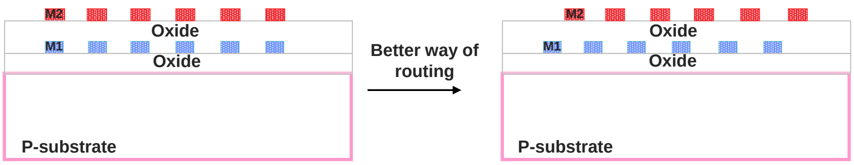
Shielding (continued)

- Even with shielding, try to route switching signals far away from very sensitive signals such as bandgap voltage or amplifier input.
- After top-level layout is done, one of the post-layout tasks is to go back and double-check the shielding on sensitive nodes.
  - Remember that coupling can happen between different layers of metal.
  - During routing, try not to put one metal route over the other.
  - For example, avoid routing metal2 exactly on top of metal1.



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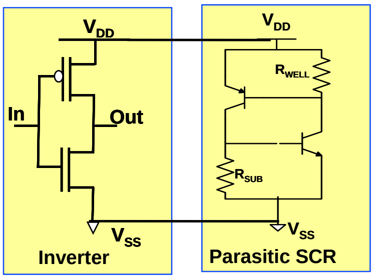
Guard Ring

One of the major concerns in CMOS processes is latchup.

- Latchup happens because of the parasitic silicon-controlled- rectifier (SCR) formation from the supply to the ground.
- To prevent latchup, R_{sub} and R_{well} should be minimized.
- A guard ring is a substrate contact around the NMOS or N_{well} contact around the PMOS.
- Adding a guard ring lowers the R_{sub} and R_{well} and prevents latchup.
- A guard ring not only helps prevent latchup but also helps to lower noise coupling through the substrate.



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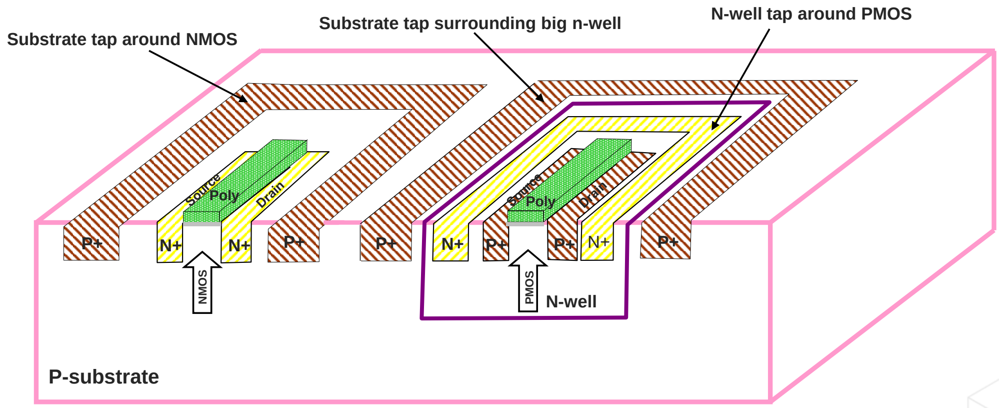
Guard Ring Guidelines

- Use substrate tap around the NMOS and connect it to the ground, especially for the large and switching NMOS.
- Use n-well tap around the PMOS and connect it to the supply, especially for large and switching PMOS.
- Surround big n-wells with substrate tap.



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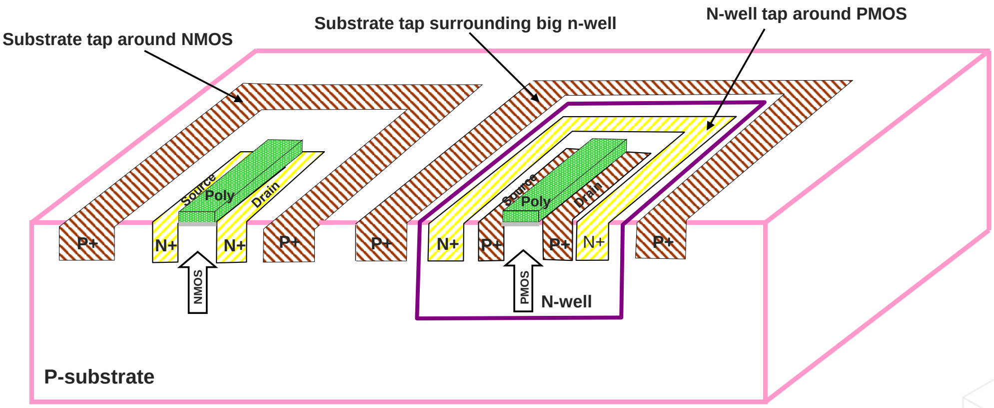
- Use substrate tap around the NMOS and connect it to the ground, especially for the large and switching NMOS.
- Use n-well tap around the PMOS and connect it to the supply, especially for large and switching PMOS.
- Surround big n-wells with substrate tap.



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Guard Ring Guidelines (continued)

- Use a double guard ring for devices where a drain or source is connected to the pad.
- Calculate metal resistance for the connection of these guard rings to the ground or supply pad.
  Use wider metal or top metal to lower this resistance.
- Use a guard ring around noisy and sensitive blocks.
- Never connect the n-well to the ground.

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Power Routing

Routing power to different blocks in the die, especially in a large die, is a challenging task in top-level routing and floorplanning.

- Power routing should be considered in floorplanning and in the block- level layout.
- Based on the power routing plan, the location and width of the power connection for each block is defined.
- Current consumption and transient-current density of each block determines the width of the supply and ground connection to that block.
- In some applications, such as regulators or power amplifiers, the power-routing consideration should be applied for the output also.
- IR drop and electromigration are main concerns for power routing.

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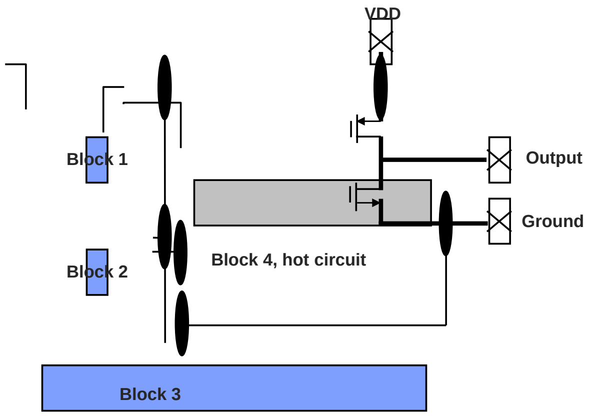
Power Routing (continued)

Hot devices need wider connections.



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Power Routing Guidelines

- Consider power routing during floorplanning.
- Before starting the layout of each block, determine the location of the block in the floorplan and define the supply and ground pin locations for the block.
- Read the rules for current-density limit for the metal, via, contact, and poly.
- Define the devices with higher current and the maximum current that they should be able to handle.
- Define the supply, ground, and other connections (such as output) of these devices.
- Based on the maximum current density and maximum current, define the width of the connection.
- Do not use top metal in block routing.

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Thermal Issues

All ICs dissipate some electrical energy that is converted to heat.

- Each new generation of ICs dissipate more heat than the previous generation because of their more demanding applications or higher operating frequencies.
- Thermal challenges for layout designer
  - Identifying the heat source
  - Identifying the heat effect
  - Managing the heat

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Heat Sources

Why is heat is generated on ICs?

- Materials are not perfect.
- Generation of heat is a consequence of fundamental electronic and electromagnetic principles.

Sources of heat in ICs

- Resistors
  - Ohmic loss
- Semiconductors
  - Ohmic loss
  - Junction loss
- High frequency losses
  - Skin effect
  - Field loss

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Heat Effect

Impacts of effect of heat on a device.

- Effect on intrinsic properties of device
  - For example, the effect of temperature on a resistor or on threshold voltage of CMOS.
  - Circuit designer should consider these effects on the design.
- Creating mechanical stress which causes change in device properties
  - For example, the effect of expansion mismatch between different parts of the IC because of the difference in metal layers or construction of these parts.
  - This is the concern of layout designer.
- Creating reliability issues
  - For example, electromigration effect becomes worse at higher temperature.

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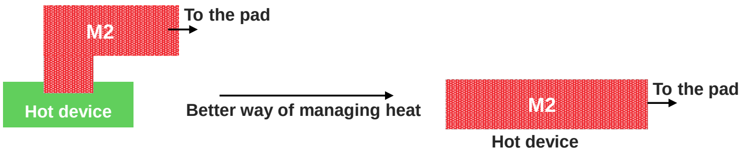
Managing Heat

- Layout designers should be aware of mechanical stress and mechanisms of heat transfer.
- The first step in managing heat is defining the heat sources (such as high-power outputs) and the thermal sensitive circuits (such as overtemperature-detection circuits).
- The metal connection to the hot devices should be symmetric with the device. Having an asymmetric metal connection around the hot device causes an expansion mismatch around it. This mismatch puts mechanical stress on the circuit.



- Heat-sensitive circuits should be placed at the center of the chip and should be symmetric with the location of the hot devices.

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Subsection

Device Orientation and Contact Sharing

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Good Design Practices

Why are these practices important?

- They have direct impact on the performance of the IC.
- They have direct effect on the reliability of the IC.
- Following these guidelines from early stage of the layout helps to have easier top-level layout.

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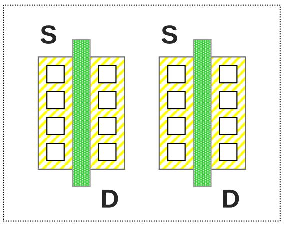
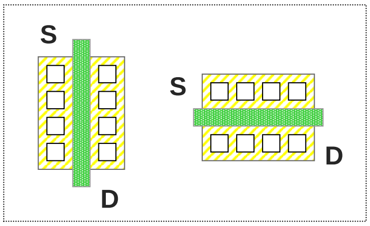
Device Orientation

The orientation of devices changes the effect of process and temperature variation.

- If two devices have the same orientation, they are affected by process and temperature variations the same way.
- If two devices have different orientations, they are affected differently by process and temperature variations.
- Matched devices should have same orientation to minimize residual mismatches between them.



Same orientation



Different orientation

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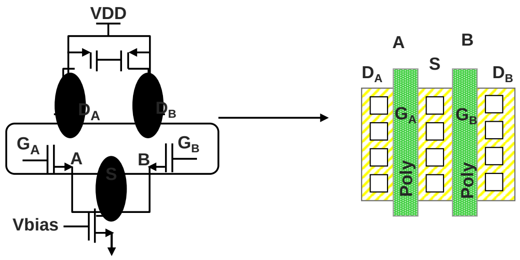
Contact Sharing

Usually, matched devices have a common source. For example, matching is required in differential amplifiers for input-differential pairs with a common source.

- For better matching and lower area, these common sources can be shared in the layout. This is called contact sharing or abutment.
- Abutment not only minimizes the layout size but also minimizes the parasitic-diffusion capacitance.



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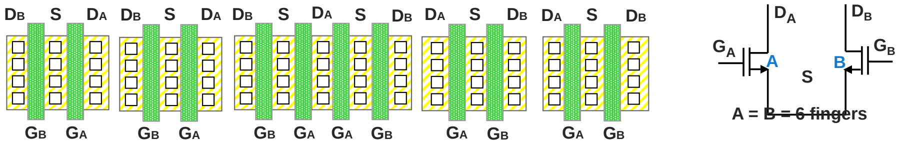
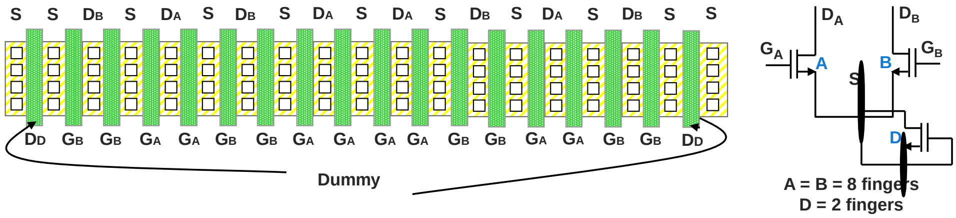
Contact Sharing (continued)

- Using contact sharing in common-centroid matching helps to make a more compact and area-efficient layout.
- Using the correct order in matching helps to have all contacts shared.



- Choosing the finger number as a multiple of four (4X) helps to cause better contact sharing for bigger devices.



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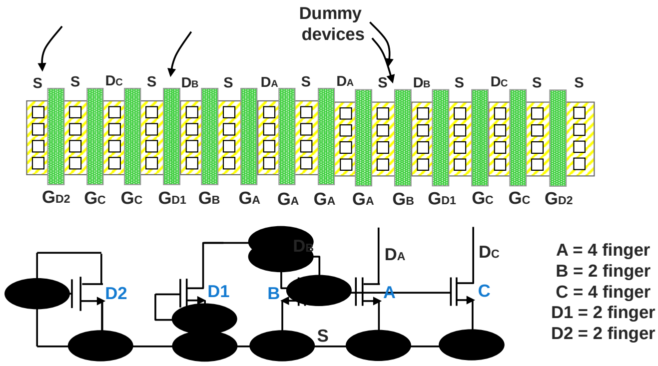
Contact Sharing (continued)

- In some circuits, such as current mirrors, the number of fingers is different for matched devices. To have a perfect common centroid with shared contacts, sometimes there should be dummy devices at the middle of matched devices.



Note: The connection of gates, drains and sources are not shown in the layout view.

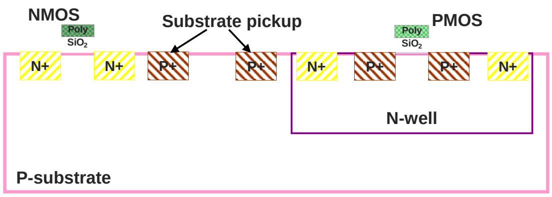
The reason for this is to be able to share all contacts. So, we need one contact for each finger of B to share their drain.

Substrate Consideration

One of the biggest concerns in layout, especially in the CMOS process, is the substrate.

- If a process is not twin-tub, the substrate voltage should be lowest across the die.
- But if a process is twin-tub, the substrate can be grounded.

  However, negative voltage can be used somewhere across the die.

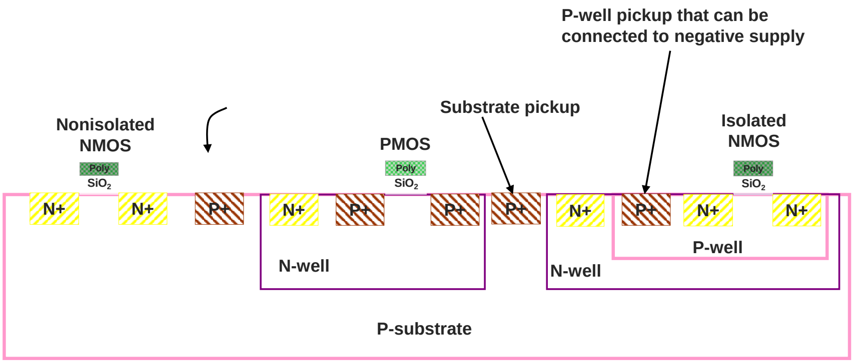


Not twin- tub process

When process is not Twin Tub the voltage of drain and/or source of NMOS can go all the way down to the lowest supply of the die. For this reason, the substrate should be connected to the lowest supply of the chip to prevent turning on parasitic diode between Drain/source and substrate.

In Twin Tub process with good design, circuit designer can use Isolated NMOS for the all NMOSs which are going to connect to lowest supply (which is usually negative supply) and the rest of the circuit (which are ground reference) can use non-isolated NMOS with substrate connected to the ground (0V).

Substrate Consideration (continued)



Twin-tub process

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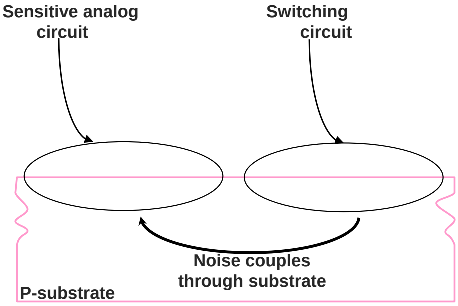
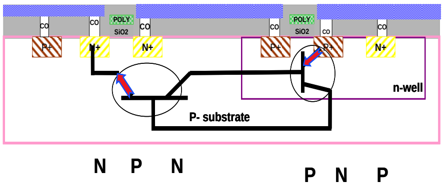
In Twin Tub process with good design, circuit designer can use Isolated NMOS for the all NMOSs which are going to connect to lowest supply (which is usually negative supply) and the rest of the circuit (which are ground reference) can use non-isolated NMOS with substrate connected to the ground (0V).

Substrate Consideration (continued)

- There are two reasons to consider substrate as an important issue:
  - Coupling of noise through substrate.
  - Formation of a latchup SCR (silicon-controlled rectifier) through the substrate.
- The grounded diffusion added around the blocks is a guard ring.



Latchup SCR forms through substrate



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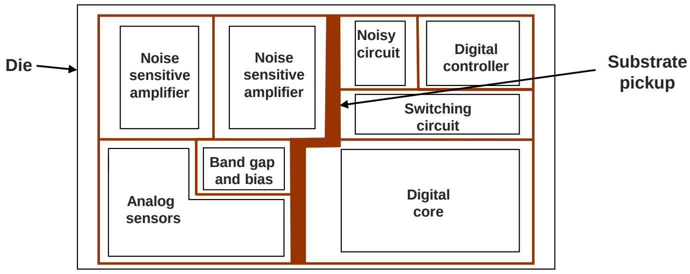
Coupling of Noise Through Substrate

Switching or noisy circuits and sensitive or analog circuits sit on the same substrate. Separating these two kind of circuits in the floorplan is one of the concerns in floorplanning.

- Like other conductive materials, the substrate has resistance.

  Typically, it is 2 kOhm/square.
- With the separation of switching and sensitive circuits, still there will be coupling between them.



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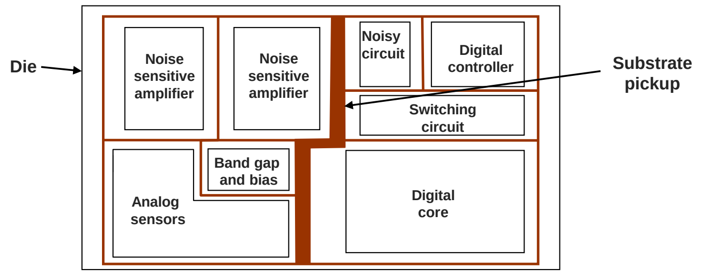
Coupling of Noise Through Substrate (continued)

- Without substrate pickup between switching or noisy circuits and sensitive or analog circuits, the noise can couple from switching to sensitive circuit.
- Putting enough pickup between these two kinds of circuits helps to couple the noise to the ground and prevent it to reach the analog circuits.



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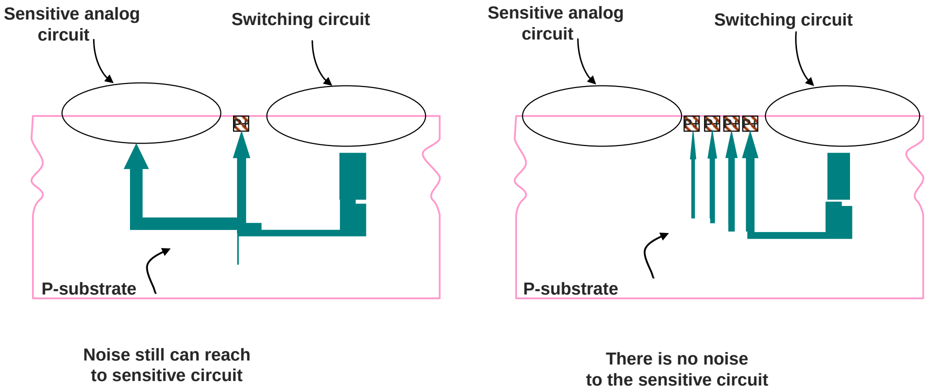
Coupling of Noise Through Substrate (continued)

- Noise coupling can happen deep in substrate.
  - Putting only one pickup between sensitive and switching circuits usually does not help much.
  - There should be more pickups to suck the noise from deep in substrate.
  - Multiple pickups help because of lowered resistance of the substrate connection.



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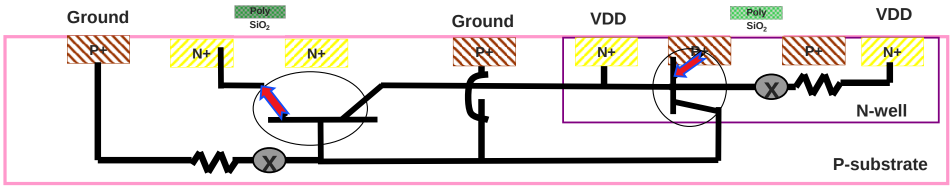
Formation of Latchup Through Substrate

Latchup is a destructive phenomena and CMOS processes are very susceptible to it.

- Normally latchup needs a trigger to inject the base current to turn on the SCR. This trigger can power up the die or some switching action somewhere in the die.
- Latchup analysis is one of the important tasks in mixed-signal design. In this analysis the susceptible circuit should be found and then, with enough substrate and n-well pickup, the loop of latchup should be broken.
- Like coupling of noise, latchup also can happen deep in substrate so multiple pickups help to reduce risk of having latchup.



Cross-sectional view

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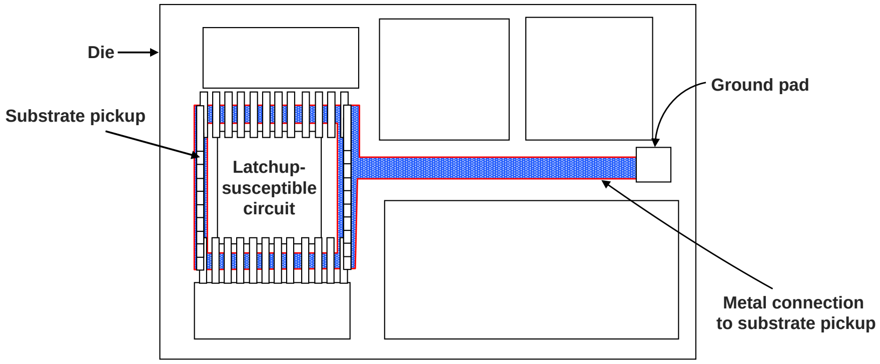
Metal Resistance of Substrate Pickup

Adding more substrate pickup is one issue and having less metal resistance is the other.

- The width of the metal connection to the substrate pickup and the distance of the pickups from the ground pad determine the resistance for the path from ground pad to the specific pickup.
- The reason for having a wide metal connection to the substrate is because of metal resistance, not because of current-density rules (requirement).



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Substrate Connection Guidelines

The device that has a drain or a source going directly to the pad is more susceptible to latchup.

- The reason is that the drain or source will be N+ (when device is NMOS) or P+ (when device is PMOS) in the SCR structure of latchup. Any signal on the pad can trigger this SCR and cause latchup.
- Put a double guard ring around such devices to prevent latchup.
- Use double (or multiple) guard rings around any big devices.

  Using a double guard ring around the device with a drain or a source connected to the pad will decrease the resistance in the SCR structure and will prevent the latchup.
- Always use substrate pickup around n-well and n-well pickup ring inside the n-well.

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Substrate Connection Guidelines (continued)

- Use a double guard ring around all blocks that have switching action, especially if this switching happens at the output of the block that goes to the pad.
- At the top level, calculate the resistance of the critical guard rings to the ground pad.
  - Die will be latchup immune if this resistance becomes lower.
  - The resistance should be less than 1-2 Ohms.
- In floorplanning always consider substrate pickups for noise and latchup purpose.

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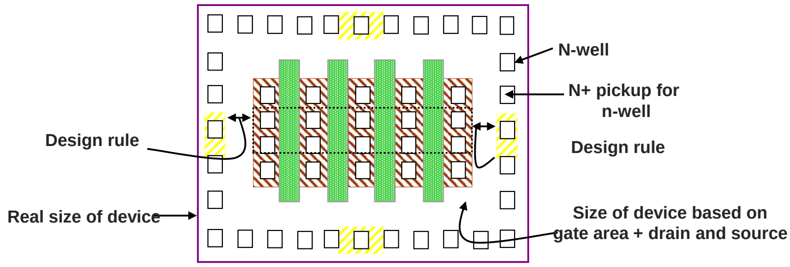
Subblock Routing

The layout of an analog or mixed-signal die starts with floorplanning.

- Designer estimates the area of each block in the die.
- This estimation is done using the design rules of the process.
- For example, only the gate area for PMOS will not be a correct estimate because of drain to n-well pickup rule.
- Merging different n-wells and p-wells as much as possible helps to reduce the size of layout.
- For complicated circuits, 20%-30% should be added because of the routing inside the block.



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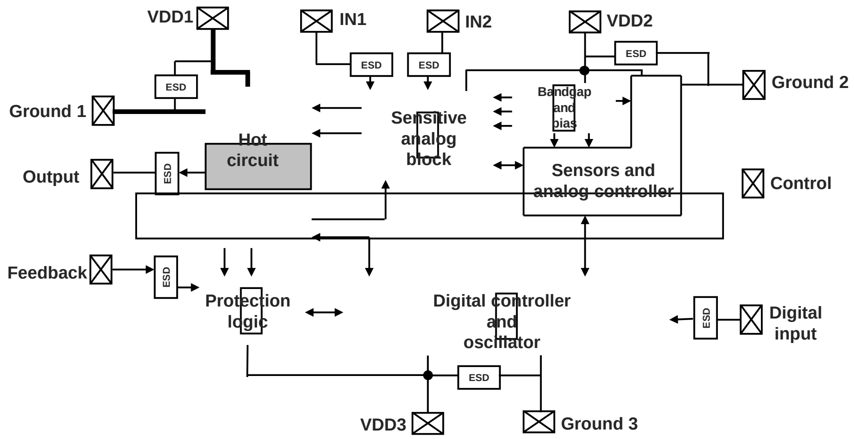
Subblock Routing (continued)

- After estimating the size of each block based on the connection between different blocks and power and output connection, floorplanning can be completed.

  Substrate pickups, ESD concerns, and connections to the pad are some of the other concerns that should be considered during floorplanning.



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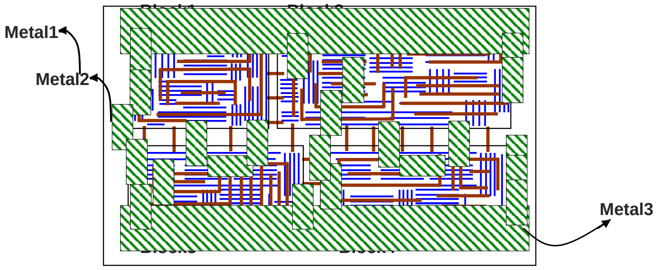
Subblock Routing (continued)

- When the floorplan is ready and the sizes are reserved for each block, the layout of subblocks can start.
- In the layout of the subblocks it is best to use a maximum of two layers of metal and leave higher metal layers for top-level routing and power-rail routing.



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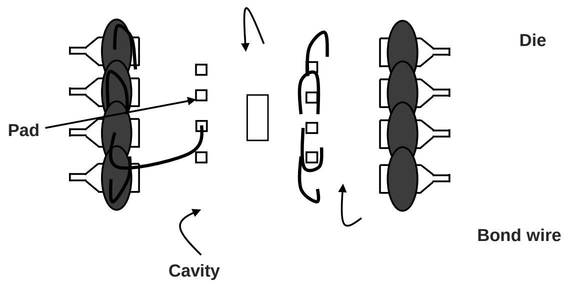
Pinout

Before starting any new project, there is a product definition which usually comes from marketing. In this product definition, the package of the final product is specified, based on the functionality and size of die.

- The first tapeout for a new product is an engineering sample that is mostly used for debugging the circuit.

  It is very useful to have the same package and pinout for the engineering sample and the final product.
- Based on the given package, the pad location can be defined on the die.



When you have same package and pin-out for engineering sample and final product some of test and reliability information can be used from engineering sample to final product.

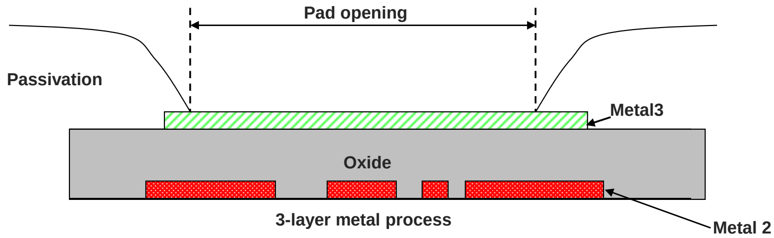
Pad

After the final step of fabrication of a die, a layer called the passivation layer covers the die to protect it. This layer does not cover some specific locations to allow bond-out. These openings on the passivation layer are called *pad opening* or *pad*.

- Based on the current-carrying capacity of each pad, the size of the pad itself and bond-wire are defined.

  For example, 1.0 mil (25.4 micron) wire can handle a maximum 1A DC current and the pad size should be at least 80 µm X 80 µm square.

- For higher current or lower inductance, a thicker bond wire, such as 1.3 mil (33.02 micron), should be used or a double- or triple-bond option should be considered.



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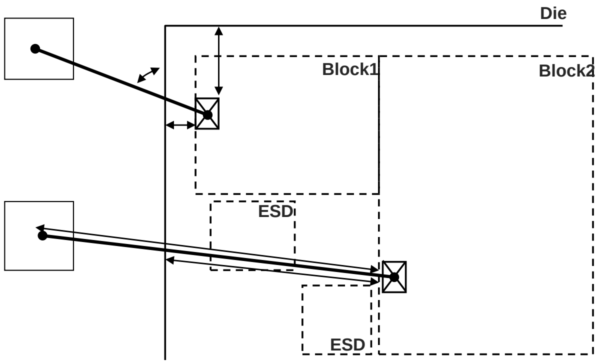
Pad Location

The location of a pad is defined by some factors.

- Packaging rules: These rules, such as maximum-wire-bond length, minimum-wire-bond angle to the edge of the die, or maximum overlap of wire bond with die are based on assembly constraints.
- The exact location of the blocks to which the specific pad is going to connect.
- The location of the ESD structure specified for the pad.



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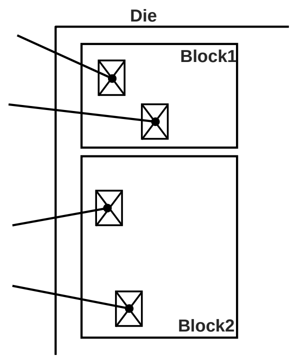
Pad-Over-Active Option

The pad-over-active option is another helpful option for choosing the appropriate pad location and pinout.

- Pad-over-active option is sometimes called circuit under pad.
- Pad-over-active option exists in some processes that allow the designer to put the circuit under the pad.
- With this option, ideally the designer can put the pad right on top of the devices that are going to be connected to it.
- Alternatively, the pad can be on top of the ESD structure.



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Pinout Selection Guidelines

With all the information about the packaging rules and pad placement options and considering the peak-current requirement of each pin and connection to the different blocks, the pin-out can be chosen.

- Remember that the preliminary pinout should be done during floor-planning, but after the layout of blocks is done.
- This pinout can be changed slightly to meet the exact requirement.

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Pinout Selection Guidelines (continued)

Besides pad and packaging rules, these are the important factors on pinout selection:

- Noise: Try to put switching and noisy pins close together and analog and sensitive pins close to each other especially at the inputs.

  If they are going to sit on the same side of the die, put VDD or ground pins between those.
- Multiple ground and VDD pad: Use multiple VDD and ground pad (if there is extra pin available in package) to have better shielding.
- Consider IR drop for power rails and high current outputs: If you have the pad-over-active option, push the pad toward the inside of the die to minimize the metal resistance of routing.
- Based on signal flow and block location, use input and output pads in different sides of the die to get more isolation from input to the output.

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Via Reliability Issues

Use vias to connect a metal layer to higher or lower metal layers.

- Each via has a certain maximum current density.
  - If a higher current passes through the via, it can damage the metal fill inside the via.
  - The damage can cause a short circuit to adjacent nodes or an open circuit in the signal path.
- The reliability concern increases with higher temperatures and in considering device lifetime.
- There is a known specification for this maximum current.

  For example, one specification is 0.4 mA per via at 110 degree Celsius.
- It is best not to use a single via. Always use more than two vias for connections.

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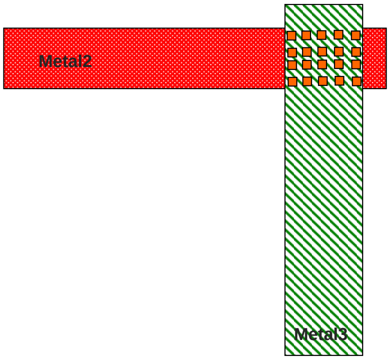
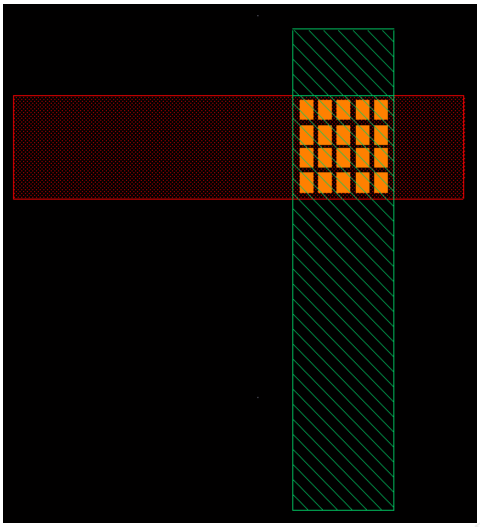
Via Reliability Issues (continued)

Define the number of vias for a junction based on the maximum current density and the maximum current that is going to pass through the junction.





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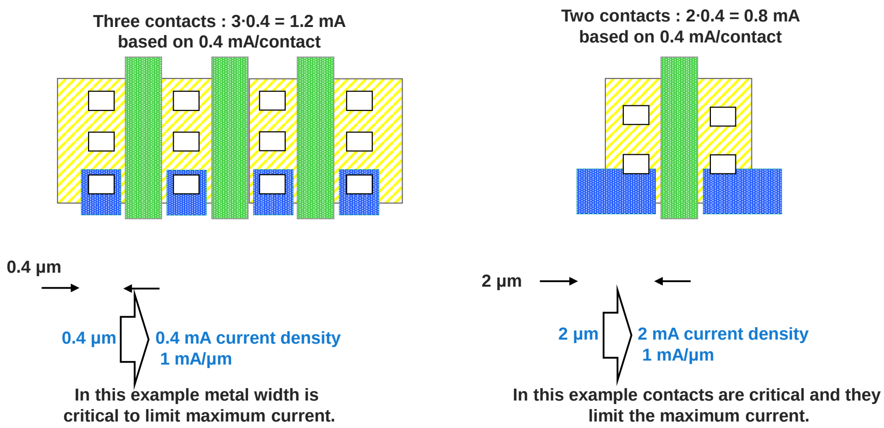
Via Reliability Issues (continued)

- Contacts, like vias, have constraints on maximum current density.
- The finger size and width of the bigger device (with high current) should be chosen based on metal and contact maximum current density.



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